GE Medical Systems

Technical Publications

Direction 2251117

Revision B

Phoenix PDU Module in 1.0/1.5T ACGD Cabinet

Operating Documentation

TABLE OF CONTENTS

IMPORTANT SAFETY INSTRUCTIONS

SAVE THESE INSTRUCTIONS — This manual contains important instructions for the ACGD Power Distribution Unit (PDU) that must be followed during installation, operation, and maintenance.

WARNING!

OPENING ENCLOSURE EXPOSES HAZARDOUS VOLTAGES. REFER SERVICE TO QUALIFIED PERSONNEL.

Note

As standards, specifications, and designs change from time to time, please ask for confirmation of the information given in this publication.

1 INTRODUCTION

1–1 General Description

The ACGD Power Distribution Unit (PDU) distributes power to various components of a General Electric MRI system while providing surge suppression and electrical isolation from the power line. Three phase input voltage is selectable among six levels: 200 Vac, 208 Vac, 380 Vac, 400 Vac, 415 Vac, and 480 Vac. Rated frequency is 50/60 Hz. Power rating is 45 kVA continuous, 54 kVA instantaneous. The PDU mounts in a standard 19–inch rack.

ACGD POWER DISTRIBUTION UNIT
ILLUSTRATION 1–1

1–2 Scope

This manual describes procedures for installation and basic operation of the ACGD Power Distribution Unit as well as procedures for failure diagnosis/isolation and installation of field replaceable components of the PDU.

2 SPECIFICATIONS

Specifications subject to revision without notice

<u>GE Part Number</u>
2238916

2–1 Electrical Specifications

<u>Power Rating</u>
45 kVA Continuous; 54 kVA Instantaneous

<u>Input Voltage</u>
200∆, 208∆, 380∆, 400∆, 415∆, or 480∆ Vac, User Selectable

<u>Frequency</u>
50/60 Hz

<u>Output Voltage</u>
208Y/120 Vac and 420∆ Vac

<u>Overload Protection</u>
Input main circuit breaker with adjustable trip settings; output distribution circuit breakers

<u>Load Current</u>
56 amperes total maximum continuous (208 V)
34 amperes total maximum continuous (420 V)

Transformer Impedance

Less than 2%

Efficiency

97% Typical

2–2 Environmental Specifications

Heat Dissipation

4606 BTU/Hr

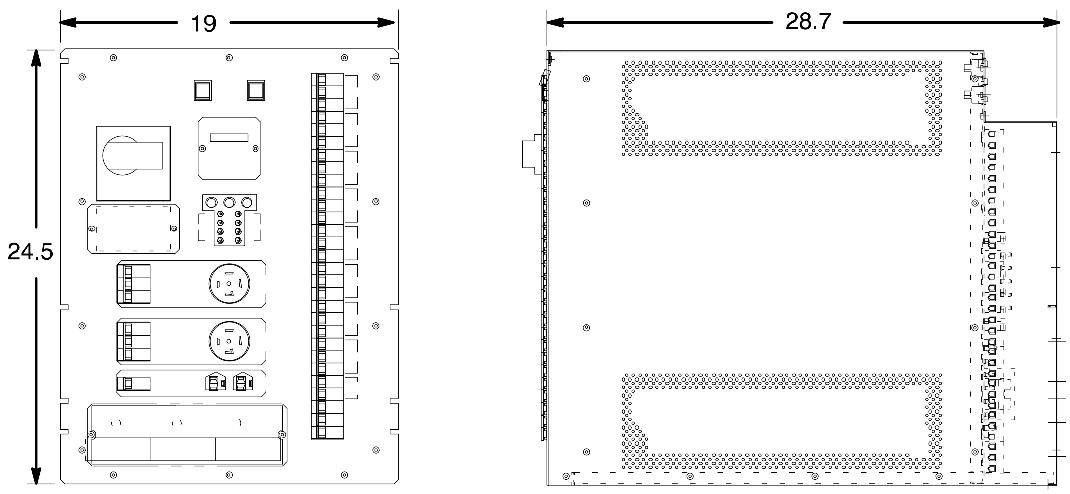
2–3 Mechanical Specifications

Weight

685 lb / 311 kg

Dimensions

See Illustration 2–1



ACGD PDU Overall Dimensions
ILLUSTRATION 2–1

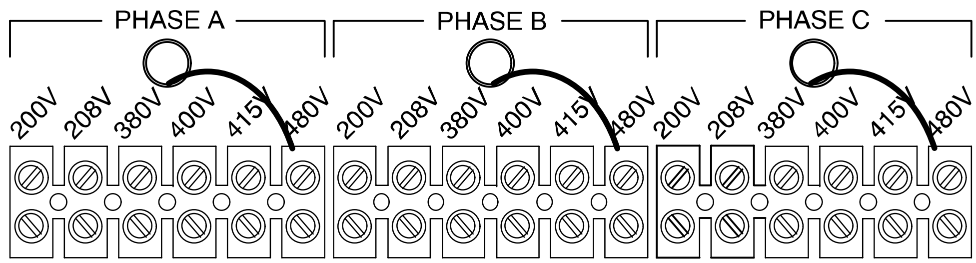
3 INSTALLATION AND OPERATION

3–1 Installation

3–1–1 Input Voltage Selection

Before connecting the input power cables, determine the nominal value of the input voltage. The PDU allows for nominal input voltages of 200, 208, 380, 400, 415, or 480 Vac (three phase). Input voltage selection is accomplished as follows:

1. On the Main Disconnect Panel (MDP), turn off the PDU circuit breaker, lock and tag.
2. Using a voltmeter or other voltage indicating device at the Main Input Terminal Board, check to be sure that no voltage is being applied to the input of the PDU.
3. Remove the plate from the front panel of the PDU marked INPUT CIRCUIT BREAKER CURRENT SELECT / INPUT VOLTAGE SELECT.
4. Illustration 3–1 shows the Input Voltage Select terminal block. For each of the three phases, be sure the wire is connected to the proper terminal for the actual input voltage at the site and that the terminal screw is tight to make a good connection.



INPUT VOLTAGE SELECT TERMINAL BLOCK
ILLUSTRATION 3–1

5. The overload and short circuit trip settings for the Input Circuit Breaker must now be set to the correct values corresponding to the input voltage.
   5a. Remove the small cover plate under the main circuit breaker which is held in place by three screws and lift open the transparent plastic cover on the lower portion of the circuit breaker to give access to the dip switches controlling the circuit breaker operation.
   5b. Illustration 3–2 shows the position of the dip switches for each input voltage selection. The switches immediately under the letter "L" are the only ones that should be changed. All others must be left in their "as shipped" up position.

3–1–2 Connection

Note the Cable Access Panel shown in Illustration 3–5 at the lower portion of the rear of the PDU. This panel provides strain relief cable clamps for the input and the various output cables. This panel may be removed from the PDU to provide easier access to the terminal blocks.

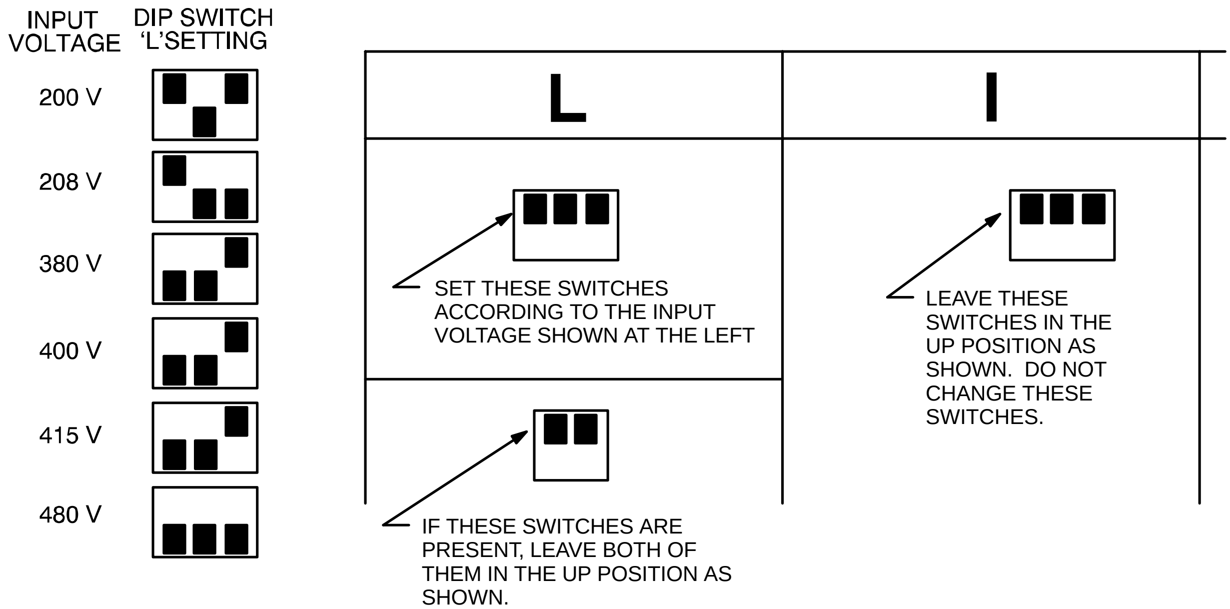
1. Route the cables through their respective cable clamps and connect cables to the appropriate terminals as shown.
2. Replace the Cable access Panel and tighten the cable clamps.

NOTE

For input cables, use wire sizes in accordance with National Electric Code Table 310–16. The following table gives maximum full–load input currents in amperes (which would occur at 10% undervoltage) for each of the several input voltages, as a guide to selecting proper wire sizes.

Maximum Input Current vs. Input Voltage
TABLE 3–1

Input Voltage	200 V	208 V	380 V	400 V	415 V	480 V
Maximum Input Current	143 A	137 A	75 A	71 A	69 A	60 A



CIRCUIT BREAKER DIP SWITCH SETTINGS
ILLUSTRATION 3–2

3–2 Operation

3–2–1 Main Input Circuit Breaker

The Main Input Circuit Breaker operating handle has three positions:

- OFF (as shown in Illustration 3–4),
- ON (vertical), and
- TRIPPED (slightly to the right of vertical). If the Main Input Circuit Breaker has been tripped, it is necessary to move the handle to the OFF position to reset it before turning it ON.

The Main Input Circuit breaker may be locked in the OFF position. By pressing on the small triangle on the operating handle, the lockout tab is made accessible to a padlock.

3–2–2 Power Off

This square pushbutton at the top of the front panel, when pressed, will immediately trip the Main Input Circuit Breaker, interrupting all power to the load.

3–2–3 EMO Reset

This square pushbutton immediately to the right of the Power Off pushbutton will enable the Emergency Stop circuit.

3–2–4 TVSS Indicators

To the right of the Main Input Circuit Breaker is a small cover plate labeled "TVSS INDICATORS." A cutout in this plate makes visible the indicator on each of the three TVSS modules. Should a TVSS module fail because of a particularly large voltage surge, a flag on the module indicator will appear showing the word "DEFECT" and the Main Input Circuit Breaker will trip. Therefore, should the Main Circuit Breaker trip unexpectedly, check the TVSS Indicators for one or more failed modules showing the word "DEFECT" (see Illustration 3–3). Any module that fails must be replaced to restore the PDU to operation with surge protection functioning. See Section 4–1–1, Replacement of a TVSS Module, for instructions for replacing a TVSS Module.

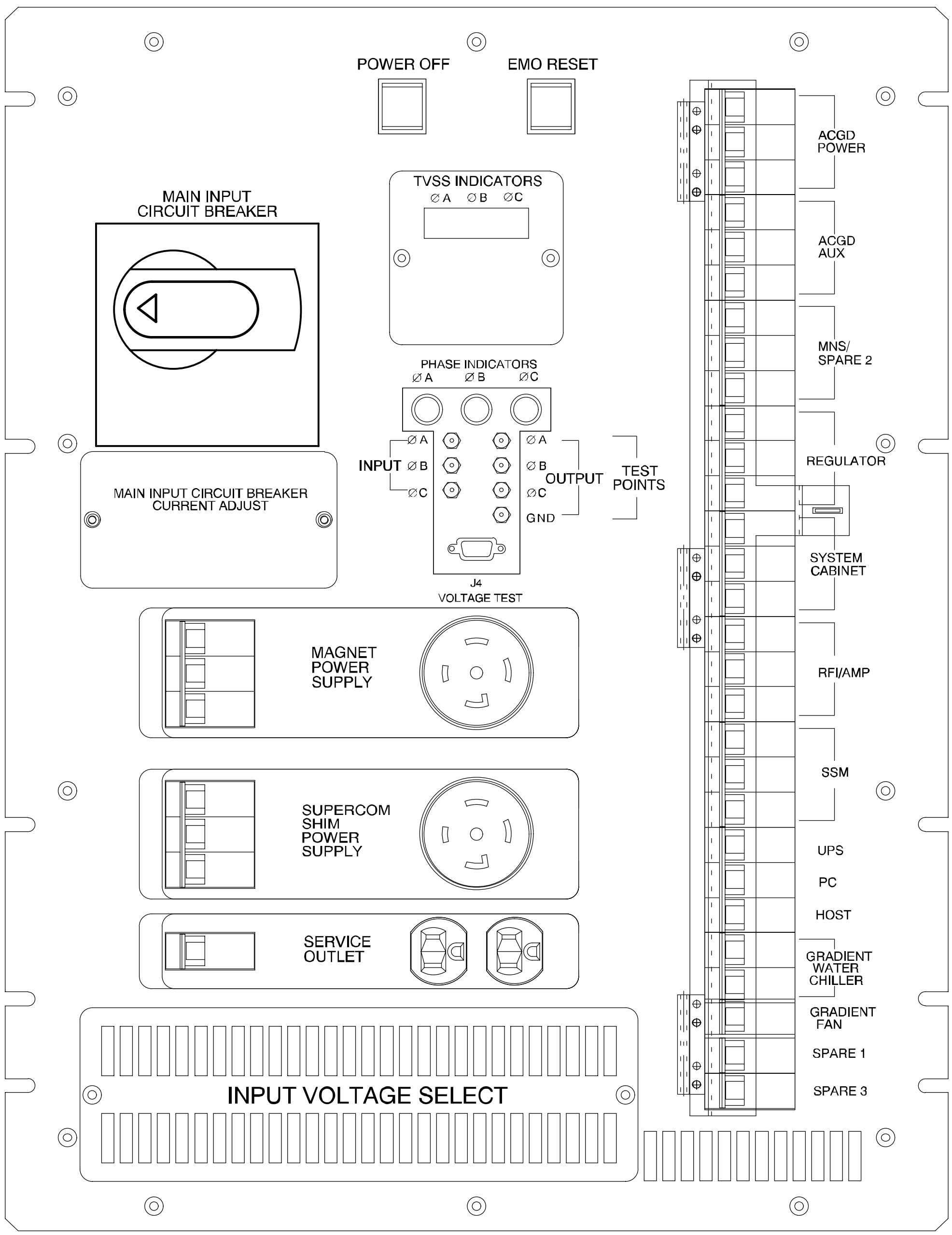
FAILED TVSS MODULE
ILLUSTRATION 3–3

3–2–5 Phase Indicators

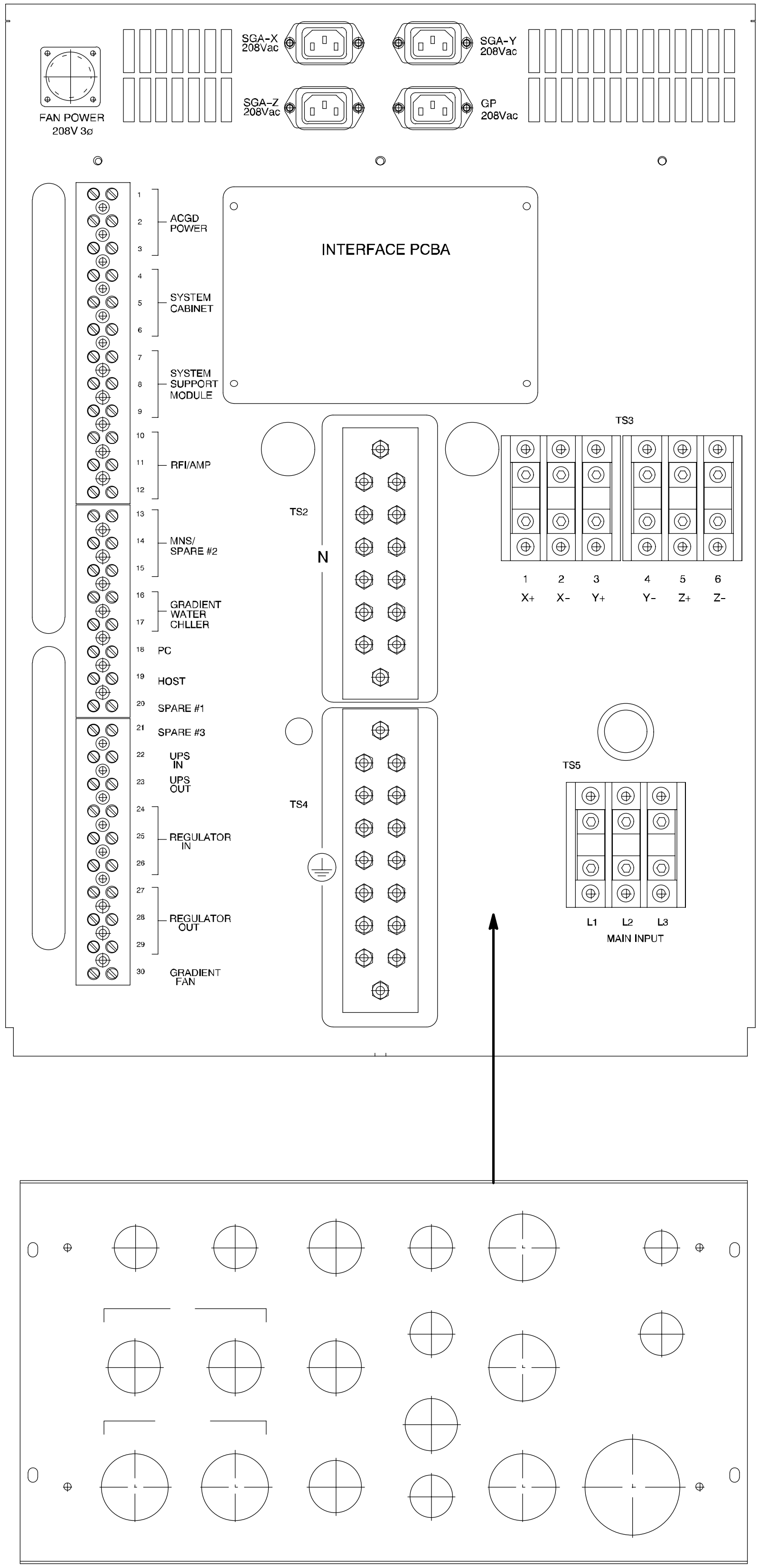
Just below the TVSS Indicator cover plate are three Phase Indicator lights. These indicate that the transformer is energized and ready to deliver power to the load.

3–2–6 Input / Output Voltage Test Points

Immediately below the Phase Incidators on the front panel are jacks into which can be plugged connectors for testing for input and output voltage. See Illustration 3–4. The voltage at these points is reduced by a factor of 100:1 from the actual voltage at the terminal blocks



FRONT PANEL COMPONENT LOCATIONS
ILLUSTRATION 3–4



REAR PANEL COMPONENT LOCATIONS

ILLUSTRATION 3–5

4 FIELD REPLACEABLE UNITS (FRU'S)

LISTING OF FRU's
TABLE 4–1

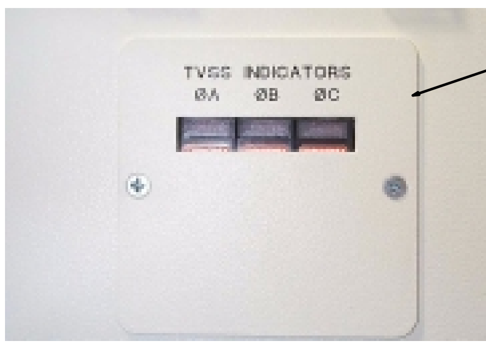
Item	I–Bus/Phoenix Part No.	Description	FRU Type
1	393–00005	TVSS Module	1

4–1 FRU Installation

4–1–1 Replacement of the TVSS Module

The operation of the TVSS Indicators is explained in Section 3–2–4, TVSS Indicators. If one (or more) of the indicators shows that a TVSS Module has failed, follow the steps below to replace the failed module and restore TVSS protection:

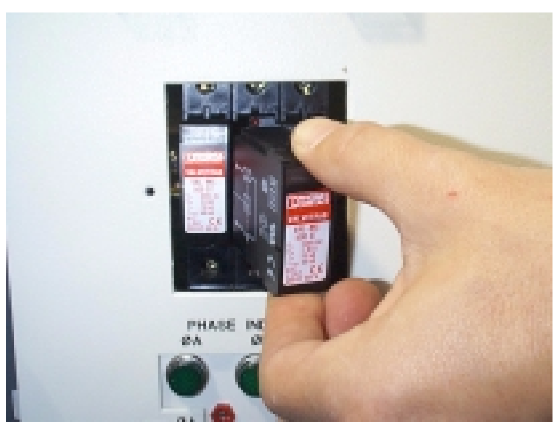
1. On the Main Disconnect Panel (MDP), turn off the PDU circuit breaker, lock and tag.
2. Using a voltmeter or other voltage indicating device, check the voltage at the input terminal block TB1 to be sure that no voltage is being applied to the input of the PDU.
3. On the front panel of the PDU, remove the small cover plate over the TVSS Modules.



TVSS COVER PLATE

TVSS COVER PLATE
ILLUSTRATION 4–1

4. Then pull out the failed TVSS Module(s) (showing the word "DEFECT") and plug in the replacement(s) as shown in Illustration 4–2.



TVSS MODULE REPLACEMENT
ILLUSTRATION 4–2

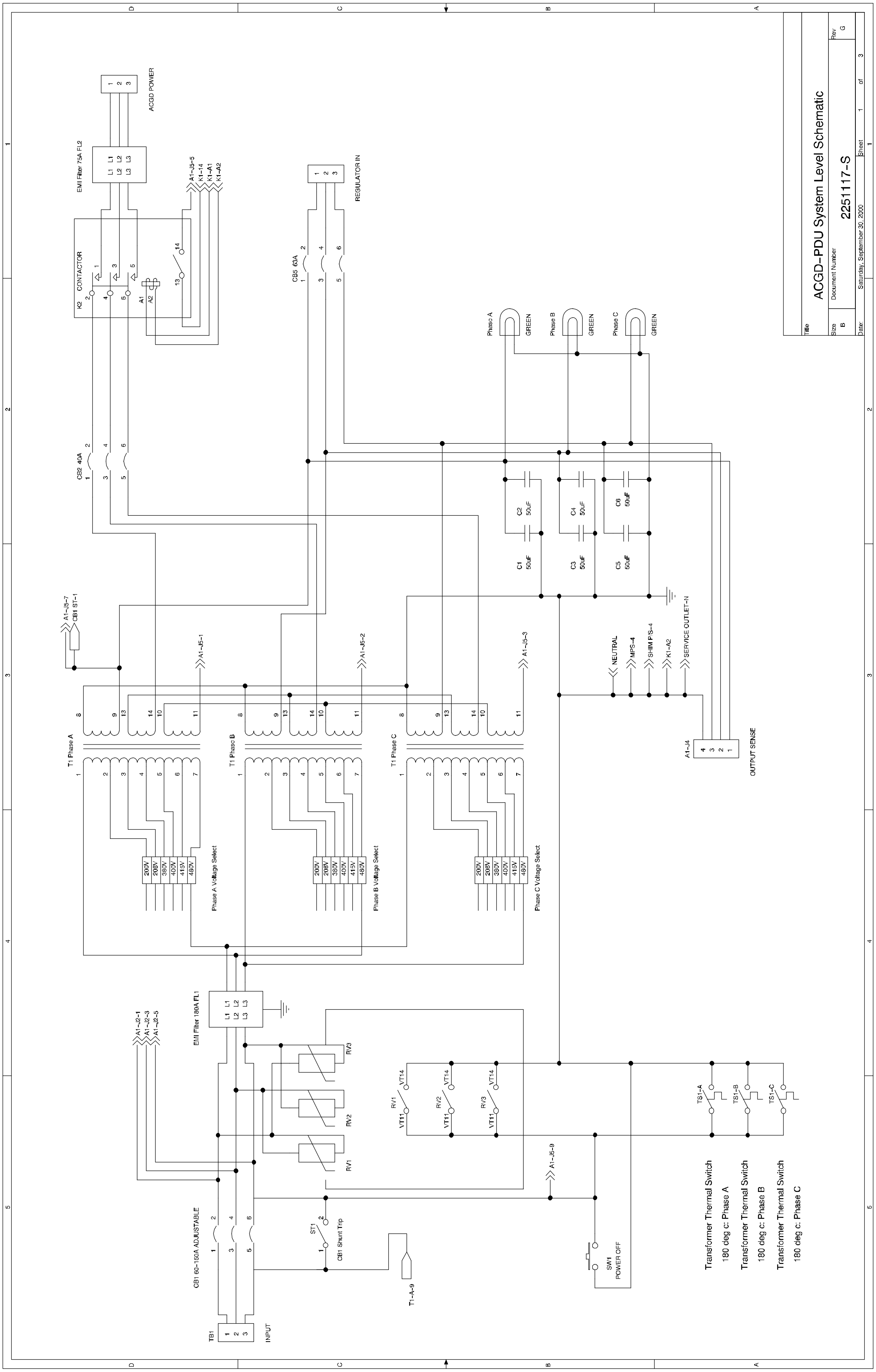
5. Replace the TVSS Cover Plate and re–energize the PDU.
6. Verify that all TVSS Modules are good (no defect displayed).

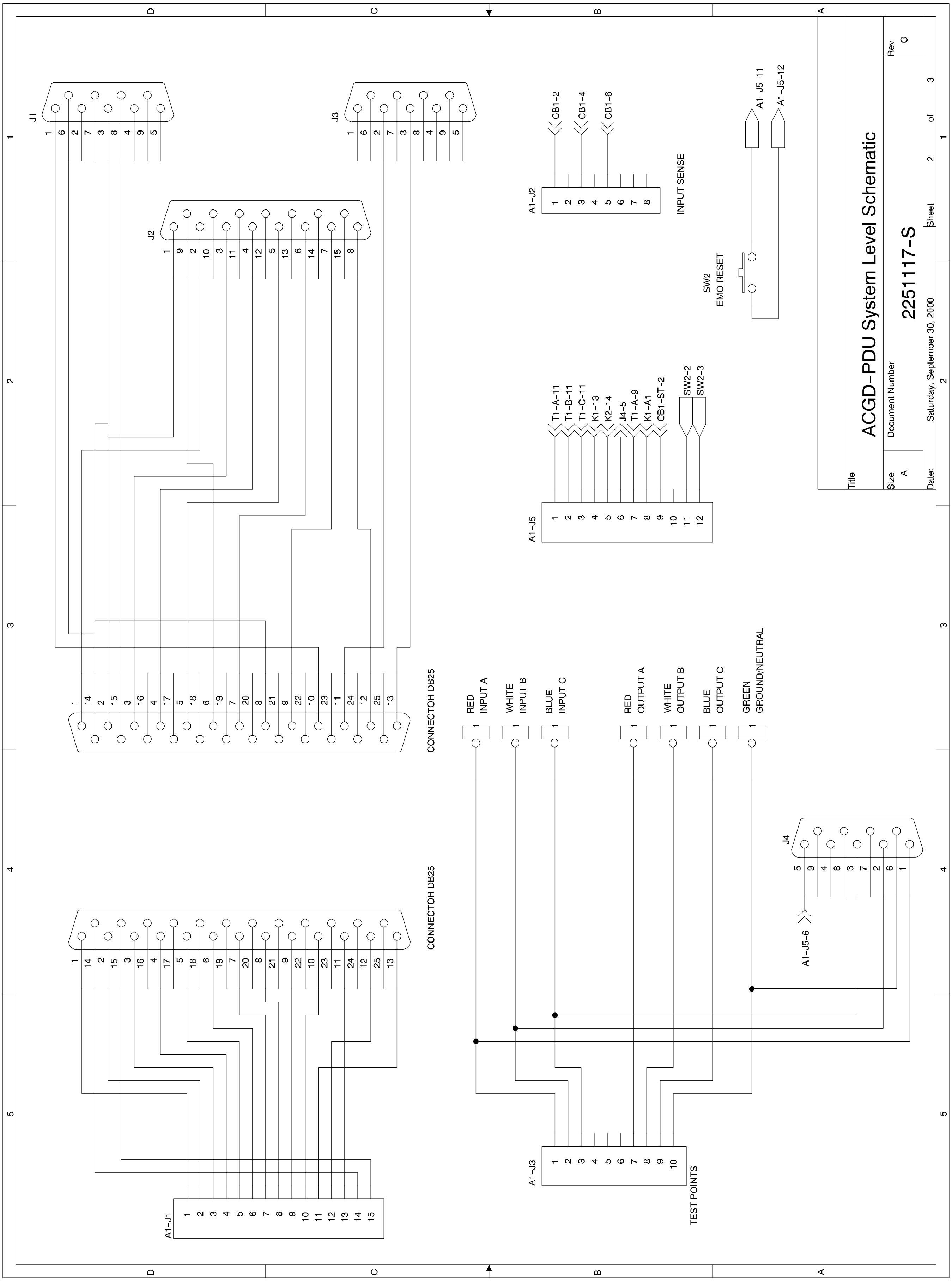
4–1–2 Troubleshooting Guide

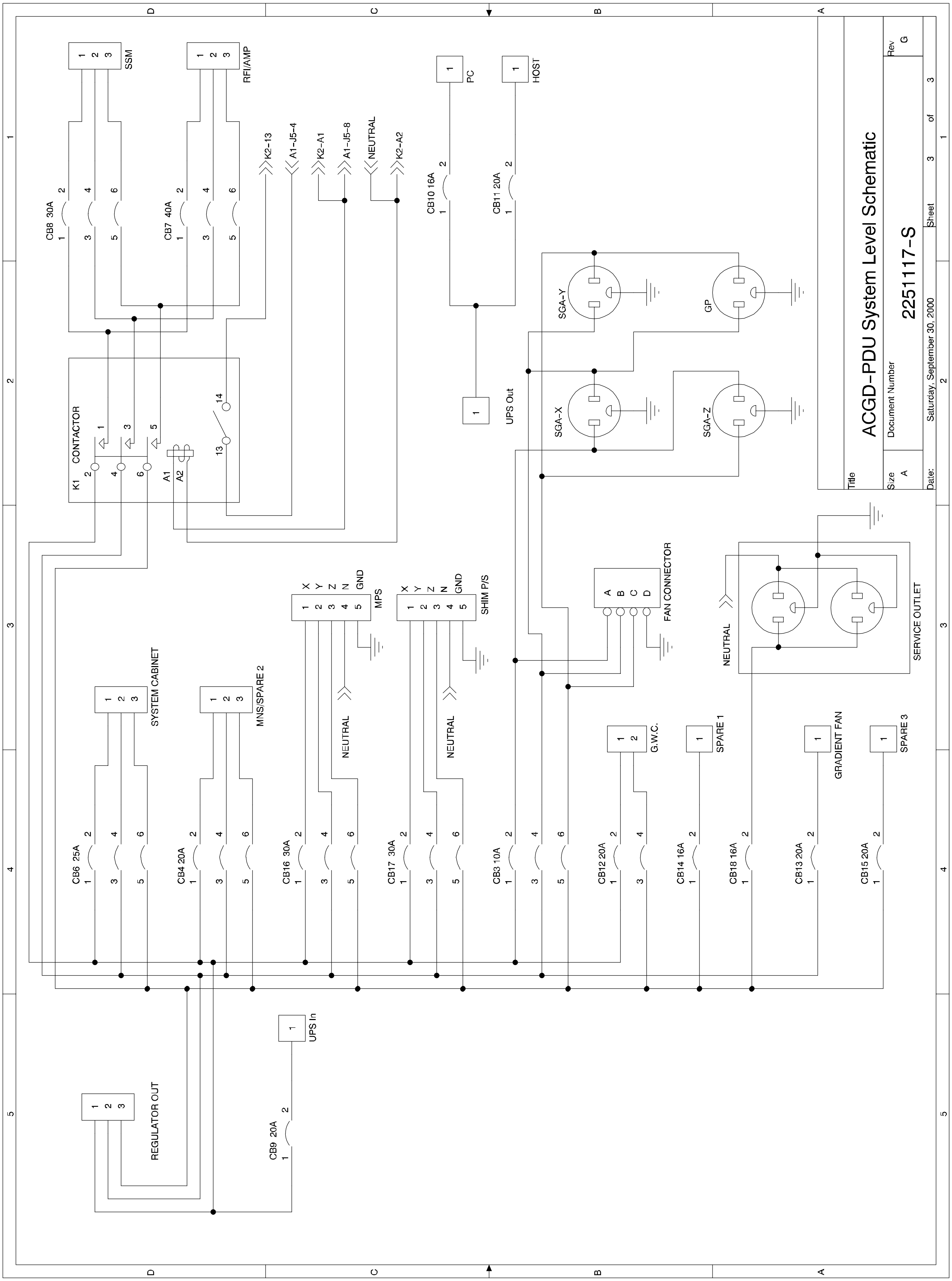
The following table lists fault conditions that may be encountered along with possible causes and solutions.

Troubleshooting Guide
Table 4–2

MODULE	SYMPTOM	CAUSE	SOLUTION
PDU	Main Input CB Trips	TVSS Module Failure	1. Check TVSS Modules for failure. “DEFECT” will appeawr in window when failure has occurred/. 2. Replace defective TVSS module.
		Overtemperature	1. Investigate cause of overtemperature 2. Allow unit to cool down. 3. Reset the circuit breaker.
		Shorted Power Off Switch	1. Turn power off. 2. Reset Main Circuit Breaker 3. Turn power on. 4. If CB trips, replace PDU
		Open from J3–2 to J3–3	1. Disconnect cable at J3 2. With DVM check for open at cable connector P3–2 to P3–3
	Distribution Breakers Tripping	Defective Circuit Breaker	1. Turn power off. 2. Check if circuit breaker still trips (won’t latch on). 3. Replace PDU
		Overload	1. Check load by disconnecting cabinet from PDU whose circuit breaker is tripping, or use and Amp Clamp / DVM to check current draw. 2. Correct load problem.
	K1/K2 Contactors not Operating	Defective contaactor	1. Ensure that the contactor is receiving 120V to A1 and A2. 2. If 120V is present, replace the contactor.
		Defective Control Board	1. Disconnect cable at J2. 2. Check at cable connector if P2–7 to P2–8 is shorted. 3. If shorted, replace PDU. 4. If not shorted, connect test plug and verify PDU operation. 5. If PDU operates with Test Plug, then replace PDU.
	Indicators not lighting when power is applied	Input circuit breaker off	1. Turn Input Circuit Breaker, CB1, on.
		Input Phase Missing	1.Check that other phase LED’s are on. 2. Replace PDU
		Light Burned Out	1.Check that other phase LED’s are on. 2. Replace PDU
		Not Getting Proper Output	1. Measure input and output, at front panel test points, of each phase, toth phase to phase (208 Vac:2.08 Vac) and phase to ground (120 Vac:1.20 Vac).. 2. If difference is not a 100:1 ratio, replace PDU.


ACGD-PDU System Level Schematic
2251117-S
Saturday, September 30, 2000
ACGD POWER
EMI Filter 75A FL2
CONTACTOR
REGULATOR IN
CB5 63A
CB2 40A
Phase A
Phase B
Phase C
GREEN
T1 Phase A
T1 Phase B
T1 Phase C
Phase A Voltage Select
Phase B Voltage Select
Phase C Voltage Select
OUTPUT SENSE
NEUTRAL
EMI Filter 180A FL1
CB1 60-150A ADJUSTABLE
CB1 Shunt Trip
SW1 POWER OFF
TB1
INPUT
Transformer Thermal Switch 180 deg c: Phase A
Transformer Thermal Switch 180 deg c: Phase B
Transformer Thermal Switch 180 deg c: Phase C


ACGD-PDU System Level Schematic
Document Number
2251117-S
Size A
Rev G
Sheet 2 of 3
Date: Saturday, September 30, 2000
J1
J2
J3
J4
A1-J2
INPUT SENSE
CB1-2
CB1-4
CB1-6
A1-J5-11
A1-J5-12
SW2
EMO RESET
A1-J5
T1-A-11
T1-B-11
T1-C-11
K1-13
K2-14
J4-5
T1-A-9
K1-A1
CB1-ST-2
SW2-2
SW2-3
CONNECTOR DB25
RED INPUT A
WHITE INPUT B
BLUE INPUT C
RED OUTPUT A
WHITE OUTPUT B
BLUE OUTPUT C
GREEN GROUND/NEUTRAL
A1-J5-6
A1-J1
A1-J3
TEST POINTS


ACGD-PDU System Level Schematic
Title
Document Number
2251117-S
Size
A
Rev
G
Date:
Saturday, September 30, 2000
Sheet 3 of 3
SSM
RFI/AMP
PC
HOST
CB8 30A
CB7 40A
CB10 16A
CB11 20A
K2-13
A1-J5-4
K2-A1
A1-J5-8
NEUTRAL
K2-A2
K1
CONTACTOR
UPS Out
SGA-Y
GP
SGA-X
SGA-Z
FAN CONNECTOR
SERVICE OUTLET
X Y Z N GND
MPS
SHIM P/S
SYSTEM CABINET
MNS/SPARE 2
G.W.C.
SPARE 1
GRADIENT FAN
SPARE 3
CB6 25A
CB4 20A
CB16 30A
CB17 30A
CB3 10A
CB12 20A
CB14 16A
CB18 16A
CB13 20A
CB15 20A
UPS In
REGULATOR OUT
CB9 20A

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
A	August 10, 2000	A. King	Initial version ACGD PDU Model 2251117 User's Manual
B	December 5, 2000	A. King	Miscellaneous revisions

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